Replication Exercise:

Structural Topic Models for Open-Ended Survey Responses

Agenda

01	Introduction	Slide 03
02	Methods	Slide 04
03	Results	Slide 05
04	Differences	Slide 06, 07
05	Replication autopsy	Slide 08
06	Extension	Slide 09

Introduction to the article

Summary

- In social science, open-ended survey responses are rarely collected and when collected, they are often analyzed using human coders.
- The authors present a semi-automatic approach known as the structural topic model (STM) that incorporates crucial information about the document which it can use as covariates like an experimental study.
- A semantically interpretable topic has two qualities: **cohesiveness** (high probability of the words occurring within documents) and **exclusivity** (top words of a topic are unlikely to appear in the top words of other topics).

Main claims of the article

- **STM classifies and helps categorize topics as good as human coders**
- **STM does not 'force' categorization, unlike the pressures on human coders who have to choose from limited options**
- **STM is not without faults or errors of its own**
  - **Misclassification of text (e.g., terror)**

Authors:
Margaret E. Roberts, University of California, San Diego
Brandon M. Stewart, Harvard University
Dustin Tingley, Harvard University
Christopher Lucas, Harvard University
Jetson Leder-Luis, California Institute of Technology
Shana Kushner Gadarian, Syracuse University
Bethany Albertson, University of Texas at Austin
David G. Rand, Yale University

Methods

- Open-ended responses from the American National Electio Survey (ANES) from 2008 were analyzed using an STM approach and compared with hand-coders of the same data, to show the effectiveness and reliability of using STMs.

- A total of 2,323 respondent's data was used.

- The respondents were asked to write about a political problem facing the United States and the two most important personal issues each of them had with regards to the election.

Results: What Was Replicable

Topic 1:
illeg, job, immigr, tax, pai, american,
care, welfar, crime, system, secur,
social, cost, health, servic, school,
languag, take, us, free

Topic 2:
immigr, illeg, legal, border, need,
worri, mexico, think, countri, law,
mexican, make, america, worker, those,
american, fine, concern, long, fenc

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Topic 1 Top Words:
        Highest Prob: immigr, illeg, legal, tax, border, pai, work
        FREX: tax, citizen, terrorist, more, amount, be, learn
        Lift: amount, tax, again, agent, between, certain, citizen
        Score: immigr, illeg, doubl, citizen, tax, pai, work
Topic 2 Top Words:
        Highest Prob: come, think, american, take, need, over, secur
        FREX: over, benefit, take, us, lower, freedom, monei
        Lift: resourc, world, afford, appli, avail, bad, believ
        Score: idea, over, take, american, come, care, mani
Topic 3 Top Words:
        Highest Prob: peopl, job, countri, here, english, worri, live
        FREX: job, worri, live, social, cost, becom, enter
        Lift: crowd, due, labor, lost, aid, anyth, assimil
        Score: job, countri, peopl, here, complex, live, worri
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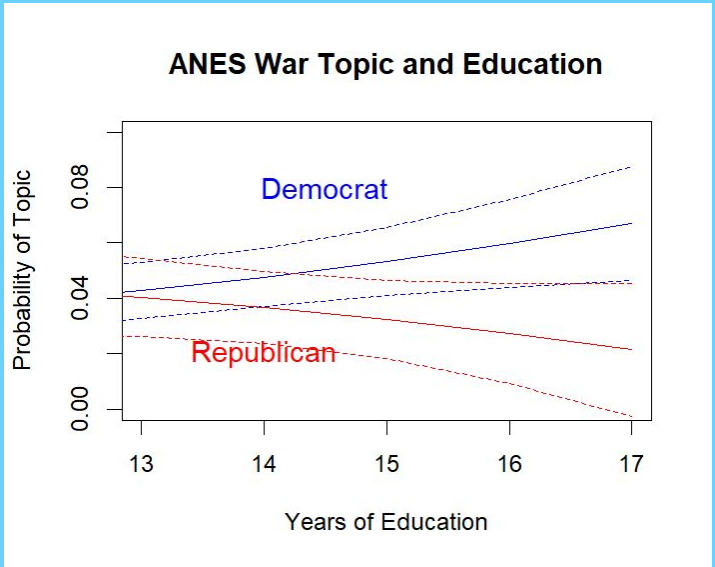
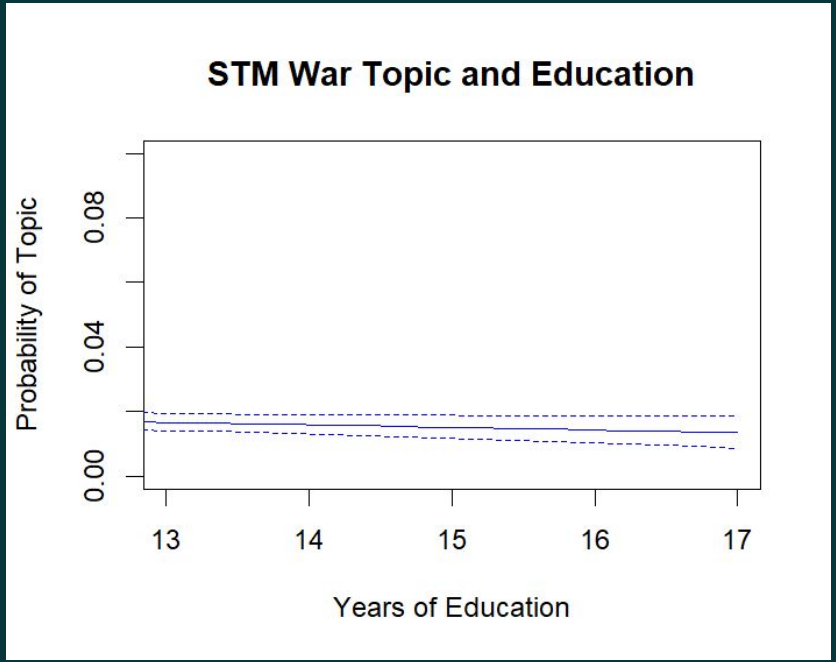
Results: What Was Replicable



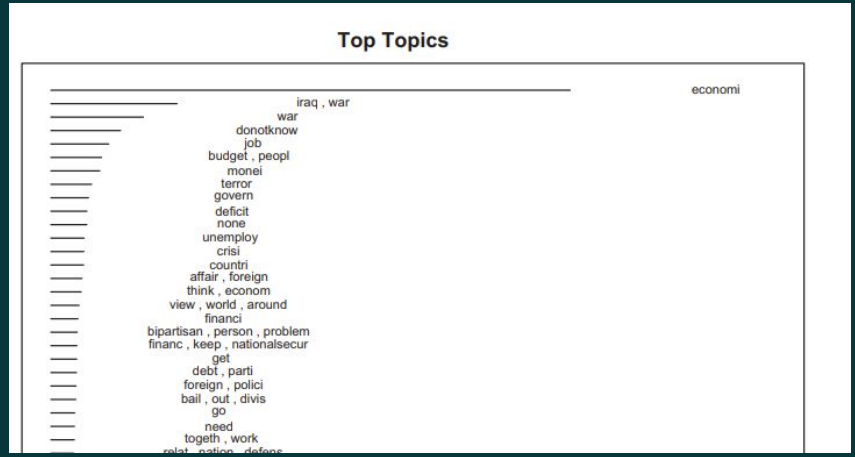
ANES, Figure 18

Differences In The Article's Results And The Replication



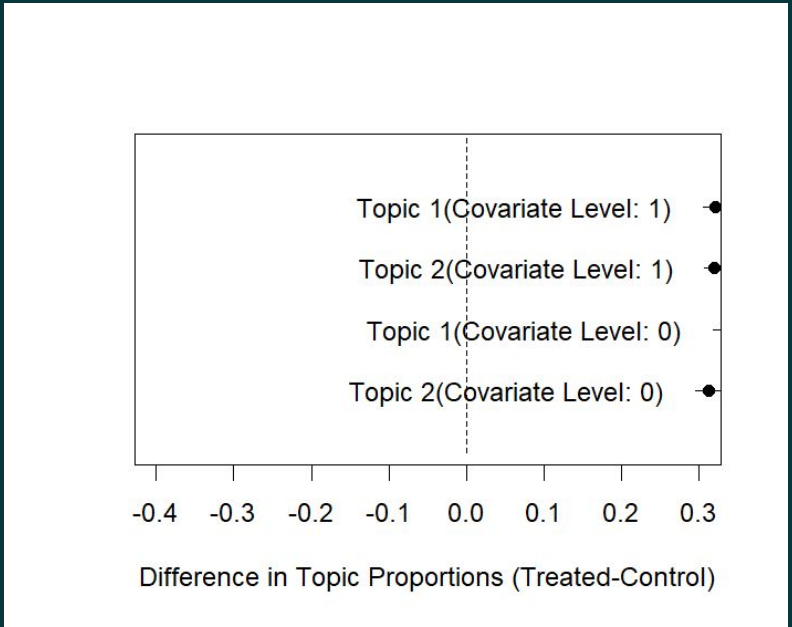
ANES, Figure 18

Differences In The Article's Results And The Replication



ANES, Figure 17

Differences In The Article's Results And The Replication



Gadarian, Figure 7

Differences In The Article's Results And The Replication

Gadarian, Figure 9

"welfare, schools, medical expenses, jobs, crime, housing, driving no insurance or license.,"

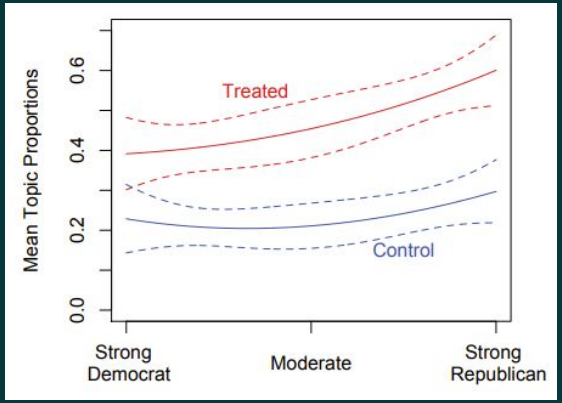
abuse
and
seem
totally
unable
to
differentiate
between
legal &
illegal
immigration

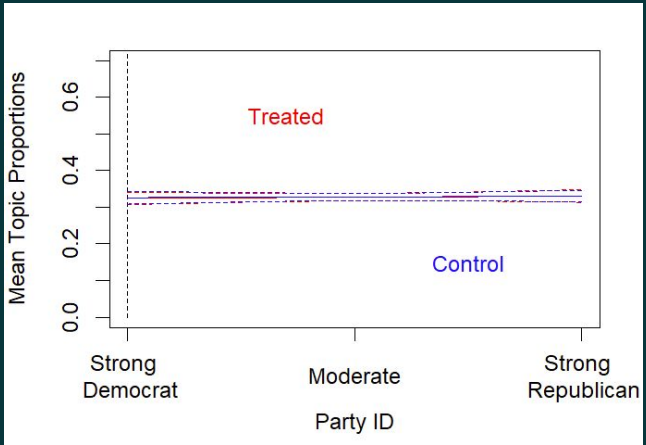
"border control, certain illegal immigrants tolerated, and others immediately deported."

to
protect
our
borders
more.
not
enough
agents
covering
too

Figure 10

Differences In The Article's Results And The Replication





Gadarian, Figure 8

Replication Autopsy

Complete Replications

Things that worked: replicating human coder's categorization for the topic of war

Partial Replications

Partially worked: Drawing of the topic that exist in the documents

Failed Replications

Where things got "stuck": Creating a STM graph mapping war and education

Extension: What Would I Do Differently

- Apply the STM outside of one social science field to determine its applicative nature
    - Cross-compare the effects of STM across disciplines to determine how it compares across subjects (e.g., is it better to use it in political texts than psychological texts)?
- Adaptability of the code and better code-book (e.g., topic ‘53’ did not match; a description in the comment would have been useful)
- Contrast STM with neural network (LLM) approaches
    - Even basic comparisons with regular dictionary approach to show the effectiveness and nuance that STM offers above other techniques, while also highlighting its weaknesses

If replicable and then furthered:

- Can explore the relationship between better estimated vocabularies and more precise treatment effects on topic proportions

Reference

Roberts, M. E., Stewart, B. M., Tingley, D., Lucas, C., Leder-Luis, J., Gadarian, S. K., Albertson, B., & Rand, D. G. (2014). Structural Topic Models for Open-Ended Survey Responses. *American Journal of Political Science*, *58*(4), 1064–1082. https://doi.org/10.1111/ajps.12103

Thank you!